source ..... mutable instruments docs
rev 2026.10

plaits

_travelers guide
_for finding your voice

24 models in 3 banks | **3** tone knobs, per-model jobs | **8** octaves on one knob

[left] press **[left]** hold › then noon = 12 o'clock cw / ccw = clockwise / counter

24 models · 3 banks · each model gives the three tone knobs and aux their own jobs (atlas)

internal lpg + envelope · trig plucks · 8-octave knob · one cv per tone knob

orange = hold

01 models

plt_01

[left]	next model in the **green** bank · pitched (wraps)
[right]	next model in the **red** bank · noise + drums
[left]+**[right]** tap	**step mode** on / off: **[left]** back · **[right]** forward through all 24 · the only way to the **yellow** bank
LED colour	yellow = new in 1.2 · green = pitched · red = noise + drums
[model] cv	moves away from the chosen model, across all 24 · steady LED = playing · slow blink = button choice
+ trig patched	model changes wait for the next trigger
power on + **[right]**	colour-blind mode: brightness instead of colour

02 pitch · tone

plt_02

[frequency]	8 octaves (c0–c8) by default · range is settable (07)
[v/oct]	–3 to +7 octaves above the knob
[fm] att	how much the fm jack bends pitch (±)
[harmonics]	spread, or the balance between the parts
[timbre]	dark · sparse → bright · dense
[morph]	sideways: variations of the same colour
cv ranges	timbre fm morph ±8 v · harmo model ±5 v · level trig 0–8 v

03 trig · level · lpg

plt_03

[trig]	fires the envelope · excites physical + drum models · plucks the lpg · samples the model cv
[level]	opens the lpg (louder + brighter) · accent on physical + drum models
trig in, cv out	**[timbre]** **[fm]** **[morph]** atts set how much internal envelope each gets
unplugging a cv	turn its att back to noon, or the envelope takes over
red 4–8	own envelope + filter · the lpg is off · level = accent
[aux]	a variant, sidekick or by-product of out

04 green bank · pitched

plt_04

1 virtual analog — two classic waves

harmo	detune between the two
timbre	narrow pulse → square → hardsync formants
morph	triangle → saw with a widening notch
aux	two hardsynced waves
tip	narrow pulse or wide notch = silence: one osc only

2 waveshaping — triangle › shaper › folder

harmo	shaper curve
timbre	fold amount
morph	asymmetry
aux	warps-style folder

3 fm — two sines, phase-modulating

harmo	frequency ratio
timbre	modulation index
morph	feedback: ccw chaotic · cw rough
aux	sub-oscillator

4 grain formant — vosim · pulsar · cz

harmo	ratio of formant 1 to 2
timbre	formant frequency
morph	formant width + shape
aux	filtered waves · harmo = lp bp hp

5 additive — 24 harmonics

harmo	number of bumps in the spectrum
timbre	loudest harmonic (like a bp cutoff)
morph	bump shape: wide → peaked
aux	hammond drawbar harmonics only

6 wavetable — 4 banks of 8×8

harmo	bank: a b c d smooth, then d c b a stepped
timbre	row (darker → brighter)
morph	column
aux	5-bit lo-fi

7 chords — four-note string / organ

harmo	chord type
timbre	inversion + transposition
morph	string-machine waves → 16-wave table
aux	root note alone

8 speech — formant · sam · lpc

harmo	vowels, then banks of words
timbre	species: daleks → chipmunks
morph	phoneme or word
aux	raw vocal cords
words	trig speaks · fm att = intonation · morph att = speed

08 LED codes

plt_08

range	chasing = lfo · one LED = c0…c7 · alternating = octaves · all lit = full
lpg settings	two rows of 4 yellow: response · decay
calibration	1st LED blinks green: send 1 v · then orange: send 3 v

05 red bank · noise + drums

plt_05

1 granular cloud — swarm of 8 saws

harmo	pitch randomness
timbre	grain density
morph	grain length · cw = merged stack
aux	sine swarm
tip	supersaw: some random, some density, full overlap

2 filtered noise — clocked noise › filter

harmo	lp → bp → hp
timbre	noise clock
morph	resonance
aux	two band-passes
pitch	frequency + v/oct track the cutoff

3 particle noise — dust › resonators

harmo	frequency randomness
timbre	particle density
morph	ccw reverb-ish all-pass · cw resonant
aux	raw dust

4 inharmonic string — mini-rings red

harmo	inharmonicity / material
timbre	exciter brightness, dust density
morph	decay
aux	raw exciter
trig	none = dust excites · 3 voices

5 modal resonator — mini-rings green

harmo	material
timbre	exciter brightness, dust density
morph	decay
aux	raw exciter
voices	1 voice, 24 partials

6 bass drum — two analog flavours

harmo	attack sharpness + overdrive
timbre	brightness
morph	decay
aux	fm triangle › sine › dirty vca
trig	none = a steady tone

7 snare drum — two analog flavours

harmo	tone ↔ noise balance
timbre	balance of the drum modes
morph	decay
aux	fm sine pair + hp noise

8 hi-hat — two analog flavours

harmo	metal ↔ noise balance
timbre	high-pass cutoff
morph	decay
aux	ring-mod pairs, clean vca

red 4–8 ignore the lpg: trig strikes them, level = accent

+ custom models · plaits palette

a free browser tool for building your own plaits firmware: pick a model for every slot (stock, braids ports, alternate takes, experimental rubato labs and community models), set the options, generate an audio-installable build.

rubato.audio/plaits-palette · by rubato labs, not mutable instruments

06 yellow bank · new in 1.2

plt_06

1 va + filter — classic waves › vcf

harmo	resonance · ccw 24 dB gentle · cw 12 dB harsh
timbre	cutoff
morph	waveform + sub level
out/aux	low-pass / 12 dB high-pass

2 phase distortion — pd / pm

harmo	distortion frequency
timbre	distortion amount
morph	asymmetry
out/aux	synced carrier (pd) / free carrier (pm)

3–5 six-op fm — 2 voices · 32 presets each

harmo	preset
timbre	modulator level
morph	stretch envelopes · no trig = time-travel
trig	alternates the 2 voices · level = velocity
banks	basses + synths · keys, plucks, perc · organs, pads, strings, brass

6 wave terrain — 8 terrains, blended

harmo	terrain · fully cw = your own
timbre	path radius
morph	path offset
out/aux	height / height as phase distortion

7 string machine — stereo filter + chorus

harmo	chord
timbre	chorus / filter amount
morph	waveform
out/aux	voices 1+3 / voices 2+4

8 chiptune — four square voices

harmo	chord
timbre	arpeggio or inversion · att = env shape
morph	pulse width / sync
out/aux	squares / nes triangle
trig	clocks the arpeggiator

reached in step mode only: tap **[left]**+**[right]**, then step

__fig.01 : hidden settings

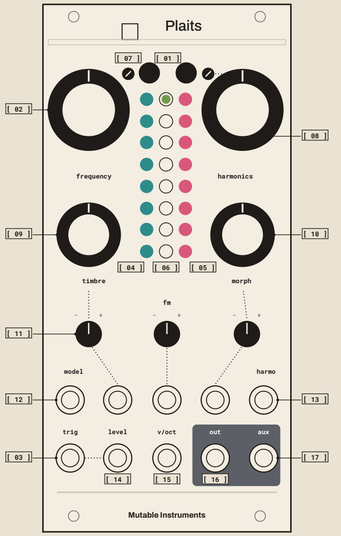
what the 8 LEDs show



saved only on the next model change

__fig.03 : front panel

01–07 = sections · 08–17 = hardware



[01] __buttons : models · banks
[02] __frequency : pitch · tone
[03] __trig : trig · level · lpg
[04] __teal icons : green bank
[05] __pink icons : red bank
[06] __LEDs : model · yellow bank
[07] __icons : held-button settings

[08] __harmonics : spread · balance
[09] __timbre : dark → bright
[10] __morph : sideways
[11] __atts : timbre · fm · morph
[12] __model : cv in ±5 v
[13] __cv ins : timbre fm morph harmo
[14] __level : lpg · accent
[15] __v/oct : –3 to +7 v
[16] __out : main
[17] __aux : variant

07 held buttons

plt_07

[left]+turn **[timbre]**	lpg response: vcfa (filter + amp) → vca
[left]+turn **[morph]**	lpg ring time + envelope decay
[right]+turn **[harmonics]**	frequency range: lfo · c0–c7 ±7 st · octaves · full 8 oct
octaves **[right]**+**[frequency]**	set the root · then **[frequency]** jumps whole octaves
[left] / **[right]** 2 s	show the lpg / range setting
saving	settings stick only after the next model change
[left]+**[right]** 2 s	calibrate: 1 v › **[left]** › 3 v › **[left]**
[timbre] jack	loads custom fm patches, a terrain or waves from the online editor